

Group77_ADL_Assignment_1

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```
[1]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler, label_binarize
from sklearn.metrics import roc_curve, auc
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.datasets import cifar10
from sklearn.metrics import confusion_matrix, accuracy_score
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
tf.random.set_seed(42)
np.random.seed(42)
RANDOM_STATE = 42
```

0.0.1 Task 1: PCA (95% energy) + Logistic Regression + ROC curves

```
[2]: # Load CIFAR-10 Dataset
(x_train_full, y_train_full), (x_test_full, y_test_full) = cifar10.load_data()

# Combine to form single dataset which can be split
X = np.vstack([x_train_full, x_test_full])
y = np.vstack([y_train_full, y_test_full]).ravel()

# Convert RGB to grayscale
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```

def rgb2gray(images):
    return np.mean(images, axis=-1, keepdims=True)

X_gray = rgb2gray(X)
n_samples, h, w, _ = X_gray.shape
print(f"Dataset: {n_samples} images, each {h}x{w} (grayscale)")

# Flatten images for PCA/logistic regression
X_flat = X_gray.reshape(n_samples, -1) # (60000, 1024)

#Train/test split 70:30
X_train, X_test, y_train, y_test = train_test_split(
    X_flat, y, train_size=0.70, random_state=RANDOM_STATE, stratify=y
)
print(f"Train: {X_train.shape[0]} images, Test: {X_test.shape[0]} images")

# Standardize features before PCA
scaler = StandardScaler(with_mean=True, with_std=True)
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Function to run PCA variant, train logistic, compute ROC curves
def run_pca_and_eval(X_tr, X_te, y_tr, y_te, pca_solver='full',
    title_suffix=''):
    # Fit PCA on training, top K components for 95% energy
    pca_full = PCA(svd_solver=pca_solver, random_state=RANDOM_STATE)
    pca_full.fit(X_tr)
    cum_energy = np.cumsum(pca_full.explained_variance_ratio_)
    K = np.searchsorted(cum_energy, 0.95) + 1 # +1 because searchsorted
    returns index
    print(f"[{pca_solver}] Components to retain 95% energy: K = {K}")

    # Fit PCA with top K components
    pca_k = PCA(n_components=K, svd_solver=pca_solver,
    random_state=RANDOM_STATE)
    X_tr_pca = pca_k.fit_transform(X_tr)
    X_te_pca = pca_k.transform(X_te)

    # visualize top principal components (eigenvectors) as images
    comps = pca_k.components_
    num_show = min(16, K)
    _, axes = plt.subplots(4, 4, figsize=(8, 8))
    axes = axes.ravel()
    for i in range(num_show):
        comp_img = comps[i].reshape(h, w)
        # normalize for display

```

```

        comp_img = (comp_img - comp_img.min()) / (comp_img.max() - comp_img.
↪min() + 1e-12)
        axes[i].imshow(comp_img, cmap='gray')
        axes[i].set_title(f"PC {i+1}")
        axes[i].axis('off')
    for j in range(num_show, 16):
        axes[j].axis('off')
    plt.suptitle(f"Top {num_show} Principal Components ({pca_solver})_
↪{title_suffix}")
    plt.show()

    # Train logistic regression classifier on PCA features
    clf = LogisticRegression(
        multi_class='multinomial', solver='saga', max_iter=2000,
↪random_state=RANDOM_STATE, n_jobs=-1
    )
    clf.fit(X_tr_pca, y_tr)

    # Compute probabilities on test set and ROC curves
    n_classes = len(np.unique(y_tr))
    y_test_binarized = label_binarize(y_te, classes=np.arange(n_classes))
    y_score = clf.predict_proba(X_te_pca)

    # Compute ROC for each class
    fpr = dict()
    tpr = dict()
    roc_auc = dict()
    for i in range(n_classes):
        fpr[i], tpr[i], _ = roc_curve(y_test_binarized[:, i], y_score[:, i])
        roc_auc[i] = auc(fpr[i], tpr[i])

    # Compute micro-average ROC curve and AUC
    fpr["micro"], tpr["micro"], _ = roc_curve(y_test_binarized.ravel(), y_score.
↪ravel())
    roc_auc["micro"] = auc(fpr["micro"], tpr["micro"])

    # Plot micro-average ROC per-class
    plt.figure(figsize=(8, 6))
    plt.plot(fpr["micro"], tpr["micro"], label=f"micro-average ROC (AUC =_
↪{roc_auc['micro']:.4f})", linewidth=2)
    # Plot curves of individual classes
    for i in roc_auc.keys():
        if i=='micro':
            continue
        plt.plot(fpr[i], tpr[i], label=f"Class {i} (AUC = {roc_auc[i]:.4f})",_
↪alpha=0.7)

```

```

plt.plot([0, 1], [0, 1], 'k--', alpha=0.4)
plt.xlim([-0.01, 1.01])
plt.ylim([-0.01, 1.01])
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title(f"ROC Curves ({pca_solver}) {title_suffix}")
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

results = {
    "K": K,
    "pca_model": pca_k,
    "clf": clf,
    "fpr": fpr,
    "tpr": tpr,
    "roc_auc": roc_auc,
    "X_tr_pca": X_tr_pca,
    "X_te_pca": X_te_pca,
}
return results

# Run full PCA
results_full = run_pca_and_eval(X_train_scaled, X_test_scaled, y_train, y_test,
    ↪pca_solver='full', title_suffix='(Full PCA)')

# Run randomized PCA
results_rand = run_pca_and_eval(X_train_scaled, X_test_scaled, y_train, y_test,
    ↪pca_solver='randomized', title_suffix='(Randomized PCA)')

# Print comparison of K and micro AUC
print("Comparison:")
print(f"Exact PCA: K = {results_full['K']}, micro AUC =
    ↪{results_full['roc_auc']['micro']:.4f}")
print(f"Randomized PCA: K = {results_rand['K']}, micro AUC =
    ↪{results_rand['roc_auc']['micro']:.4f}")

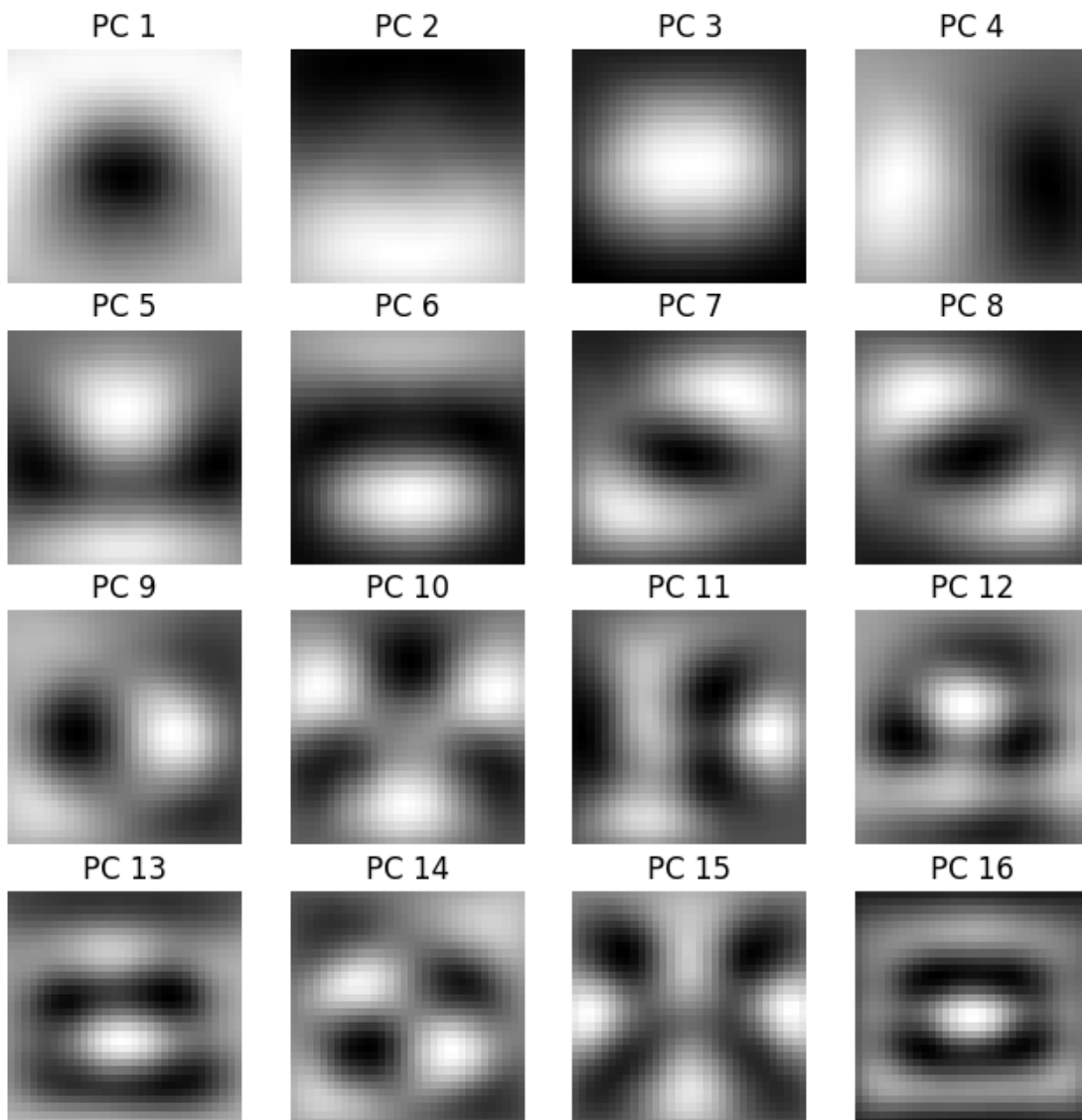
# Plot both micro ROC curves on the same figure for direct comparison
plt.figure(figsize=(8,6))
plt.plot(results_full['fpr']['micro'], results_full['tpr']['micro'],
    ↪label=f"Exact PCA micro AUC {results_full['roc_auc']['micro']:.4f}",
    ↪linewidth=2)
plt.plot(results_rand['fpr']['micro'], results_rand['tpr']['micro'],
    ↪label=f"Randomized PCA micro AUC {results_rand['roc_auc']['micro']:.4f}",
    ↪linewidth=2, linestyle='--')
plt.plot([0,1],[0,1], 'k--', alpha=0.3)

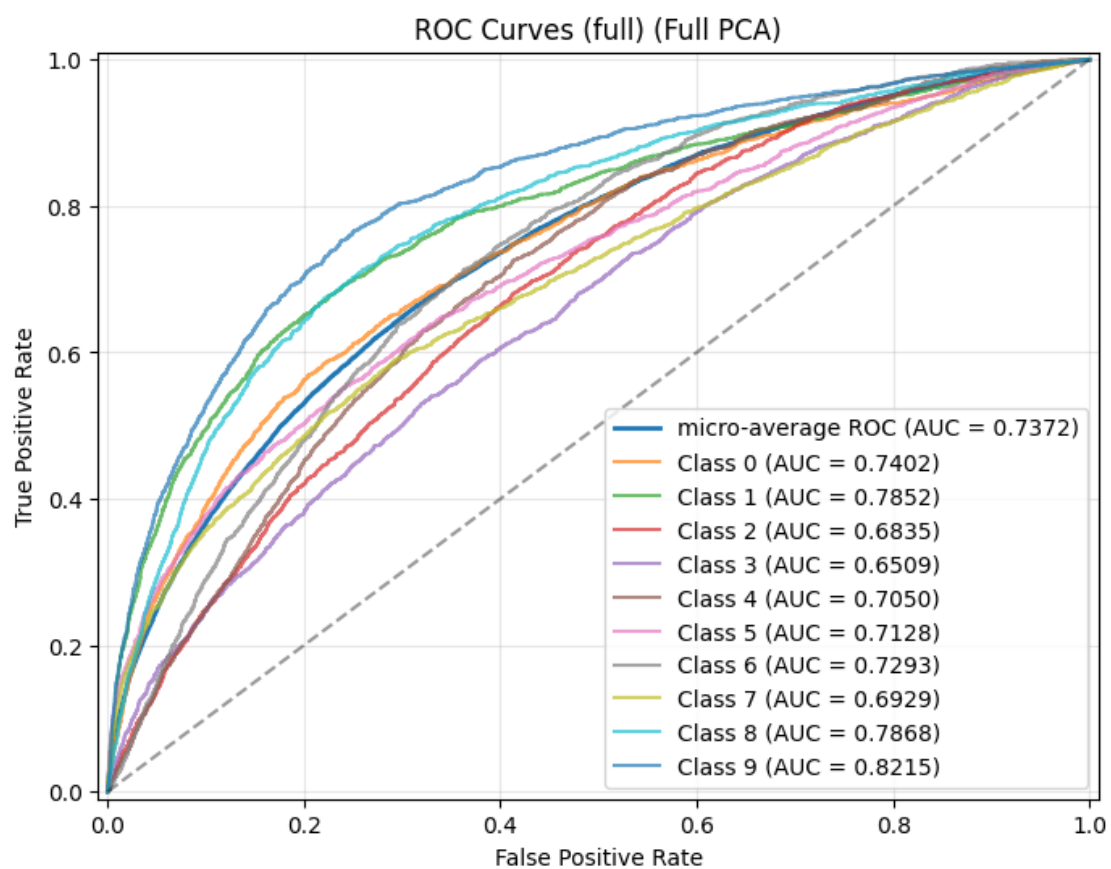
```

```
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Micro-average ROC: Exact PCA vs Randomized PCA")
plt.legend(loc='lower right')
plt.grid(alpha=0.3)
plt.show()
```

Dataset: 60000 images, each 32x32 (grayscale)
 Train: 42000 images, Test: 18000 images
 [full] Components to retain 95% energy: $K = 160$

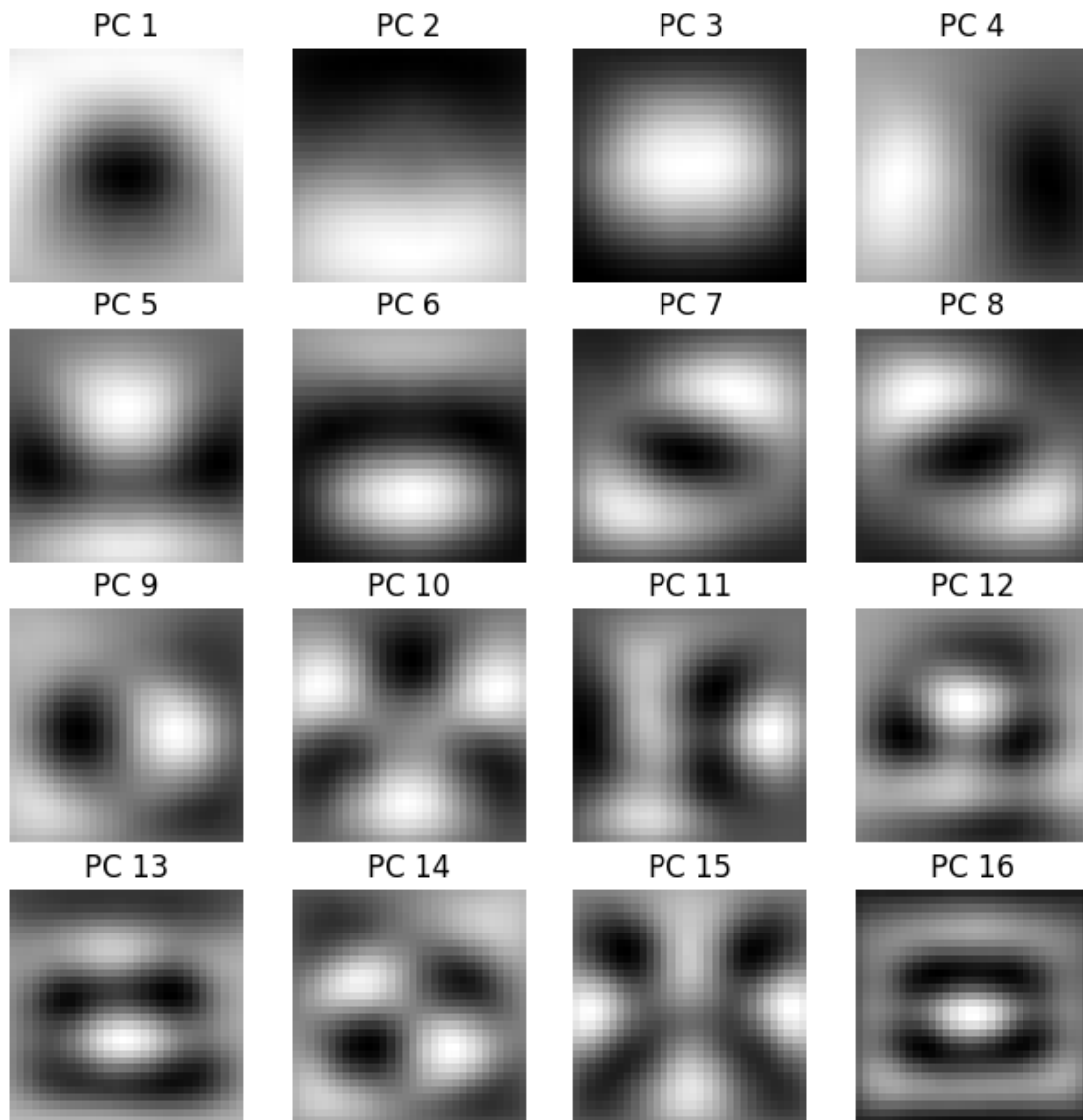
Top 16 Principal Components (full) (Full PCA)

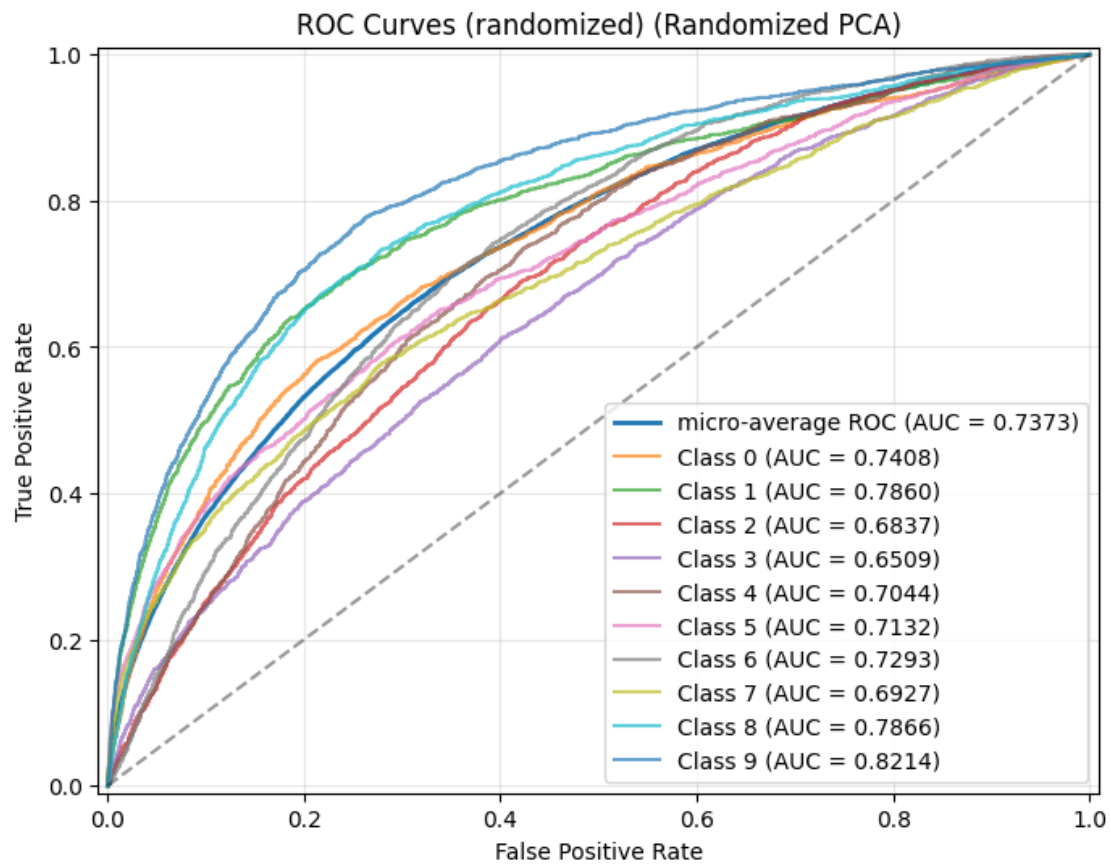




[randomized] Components to retain 95% energy: $K = 160$

Top 16 Principal Components (randomized) (Randomized PCA)

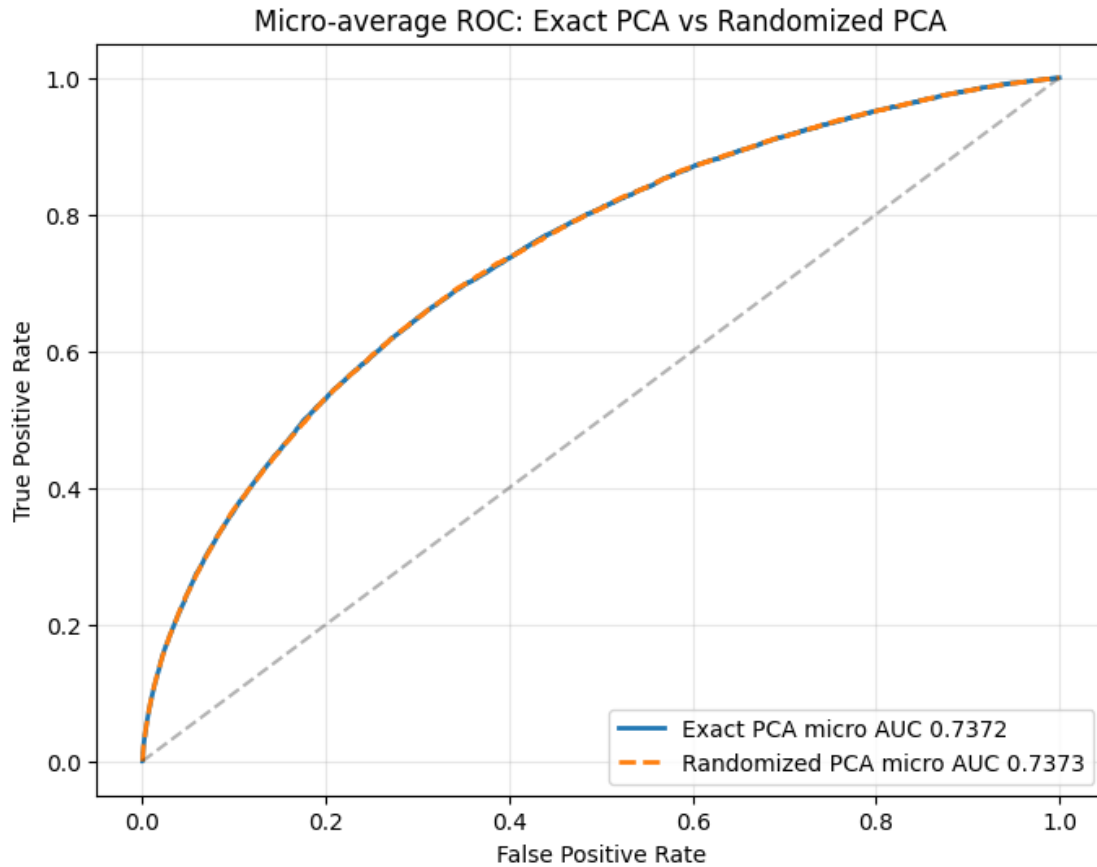




Comparison:

Exact PCA: $K = 160$, micro AUC = 0.7372

Randomized PCA: $K = 160$, micro AUC = 0.7373



0.0.2 Task 2: Single-layer linear autoencoder (K encoder nodes) with tied weights and unit-magnitude weight vectors

```
[3]: # PCA from task 1
pca_k = results_full['pca_model']
K_components = results_full['K']
M_display = min(16, K_components)

def show_components(components, title, n_show=M_display, img_shape=(h, w)):
    n_show = min(n_show, components.shape[0])
    cols = 4
    rows = int(np.ceil(n_show / cols))
    fig, axes = plt.subplots(rows, cols, figsize=(cols*2.5, rows*2.5))
    axes = axes.ravel()
    for i in range(len(axes)):
        axes[i].axis('off')
    for i in range(n_show):
        comp = components[i].reshape(img_shape)
        # normalize for display
```

```

        comp = (comp - comp.min()) / (comp.max() - comp.min() + 1e-12)
        axes[i].imshow(comp, cmap='gray')
        axes[i].set_title(f"{title} {i+1}")
    plt.suptitle(title)
    plt.show()

show_components(pca_k.components_, "PCA top components (eigenvectors)")

# Build linear autoencoder with tied weights: encoder  $W$  (input_dim  $\rightarrow$  K),
#  $\hookrightarrow$  decoder uses  $W^T$ 
input_dim = X_train_scaled.shape[1]
K_latent = K_components # use same latent dimension as PCA

print(f"Building linear autoencoder with input_dim={input_dim}, latent_
 $\hookrightarrow$  K={K_latent}")

class TiedLinearAutoencoder(keras.Model):
    def __init__(self, input_dim, latent_dim):
        super().__init__()
        w_init = tf.random.normal(shape=(input_dim, latent_dim), stddev=0.01,
 $\hookrightarrow$  seed=RANDOM_STATE)
        self.W = tf.Variable(w_init, trainable=True, dtype=tf.float32,
 $\hookrightarrow$  name="W_tied")
        self.input_dim = input_dim
        self.latent_dim = latent_dim

    def call(self, inputs):
        # inputs: (batch, input_dim)
        # encoder:  $z = X @ W \rightarrow$  shape (batch, latent_dim)
        z = tf.linalg.matmul(inputs, self.W)
        # decoder:  $X_{hat} = z @ W^T \rightarrow$  (batch, input_dim)
        x_hat = tf.linalg.matmul(z, tf.transpose(self.W))
        return x_hat

    def get_weights_matrix(self):
        return self.W.numpy()

# Create model
ae = TiedLinearAutoencoder(input_dim=input_dim, latent_dim=K_latent)

# Loss and optimizer
mse_loss_fn = tf.keras.losses.MeanSquaredError()
optimizer = tf.keras.optimizers.Adam(learning_rate=1e-3)

# Renormalize columns of  $W$  to unit norm
@tf.function

```

```

def renormalize_columns(W):
    # unit L2 norm for each column
    col_norms = tf.norm(W, axis=0, keepdims=True)
    # avoid division by zero
    col_norms = tf.where(col_norms < 1e-12, tf
                        .ones_like(col_norms), col_norms)
    W_normalized = W / col_norms
    return W_normalized

# Training step
@tf.function
def train_step(model, x_batch):
    with tf.GradientTape() as tape:
        x_hat = model(x_batch)
        loss = mse_loss_fn(x_batch, x_hat)
        grads = tape.gradient(loss, [model.W])
        optimizer.apply_gradients(zip(grads, [model.W]))
        # Enforce tied-weights constraint (decoder uses  $W^T$  automatically) and
        ↪ unit-magnitude columns
        W_new = renormalize_columns(model.W)
        model.W.assign(W_new)
    return loss

# Training loop
batch_size = 256
epochs = 50
train_dataset = tf.data.Dataset.from_tensor_slices(X_train_scaled.astype(np.
    ↪ float32)).shuffle(10000, seed=RANDOM_STATE).batch(batch_size)

print("Start training linear autoencoder...")
for epoch in range(1, epochs + 1):
    epoch_losses = []
    for step, x_batch in enumerate(train_dataset):
        loss_val = train_step(ae, x_batch)
        epoch_losses.append(loss_val.numpy())
    mean_loss = np.mean(epoch_losses)
    if epoch % 5 == 0 or epoch == 1:
        print(f"Epoch {epoch}/{epochs} - train MSE: {mean_loss:.6f}")
print("Training finished.")

# Evaluate reconstruction error on test set
X_test_tf = tf.convert_to_tensor(X_test_scaled.astype(np.float32))
X_test_hat = ae(X_test_tf).numpy()
test_mse = np.mean((X_test_scaled - X_test_hat)**2)
print(f"Autoencoder reconstruction MSE on test set: {test_mse:.6f}")

# Extract weight vectors and visualize them side-by-side with PCA eigenvectors

```

```

W_final = ae.get_weights_matrix()

# W_final columns correspond to encoder's basis vectors; reshape each column to
↳image for display
W_components = W_final.T
print("W_components shape (for display):", W_components.shape)

# Show side-by-side for top M_display components: PCA_i vs AE_weight_i
n_show = min(M_display, K_latent)
cols = 2 # PCA | AE
rows = n_show
fig, axes = plt.subplots(rows, cols, figsize=(cols*3.5, rows*2.5))
for i in range(n_show):
    # PCA component i
    pca_comp = pca_k.components_[i].reshape((h, w))
    pca_comp = (pca_comp - pca_comp.min()) / (pca_comp.max() - pca_comp.min() +
↳1e-12)
    axes[i, 0].imshow(pca_comp, cmap='gray')
    axes[i, 0].axis('off')
    axes[i, 0].set_title(f"PCA PC {i+1}")
    # AE weight column i:
    ae_comp = W_components[i].reshape((h, w))
    ae_comp = (ae_comp - ae_comp.min()) / (ae_comp.max() - ae_comp.min() +
↳1e-12)
    axes[i, 1].imshow(ae_comp, cmap='gray')
    axes[i, 1].axis('off')
    axes[i, 1].set_title(f"AE weight {i+1}")
plt.suptitle("PCA eigenvectors (left) vs Autoencoder weight vectors (right)")
plt.tight_layout()
plt.show()

# Numeric comparison: correlation between PCA components and AE weight vectors
↳based on cosine similarity
def cosine_sim(a, b):
    a = a.flatten()
    b = b.flatten()
    a = a / (np.linalg.norm(a) + 1e-12)
    b = b / (np.linalg.norm(b) + 1e-12)
    return float(np.dot(a, b))

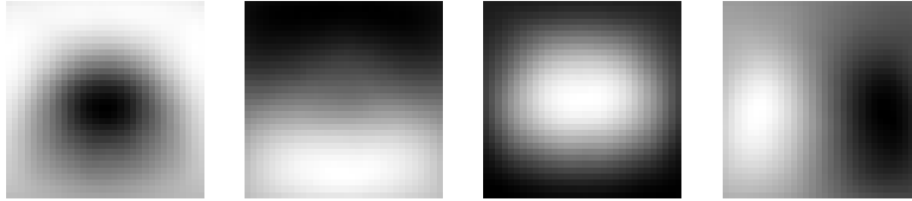
sims = []
for i in range(n_show):
    sim = cosine_sim(pca_k.components_[i], W_components[i])
    sims.append(sim)
    print(f"Cosine similarity PCA_{i+1} vs AE_{i+1}: {sim:.4f}")

print("Average cosine similarity (top shown):", np.mean(sims))

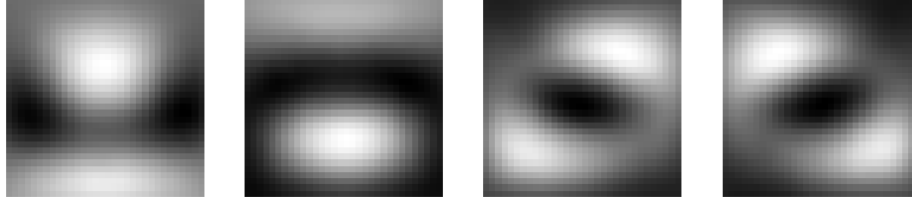
```

PCA top components (eigenvectors)

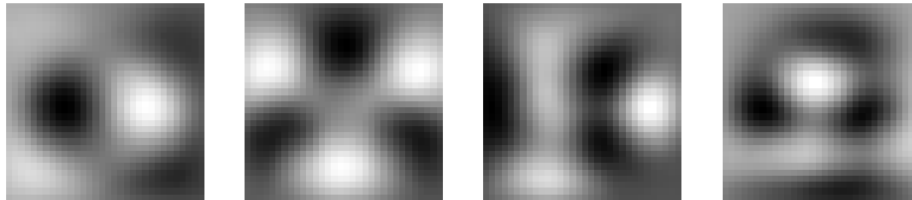
PCA top components (eigenvectors) 1 PCA top components (eigenvectors) 2 PCA top components (eigenvectors) 3 PCA top components (eigenvectors) 4



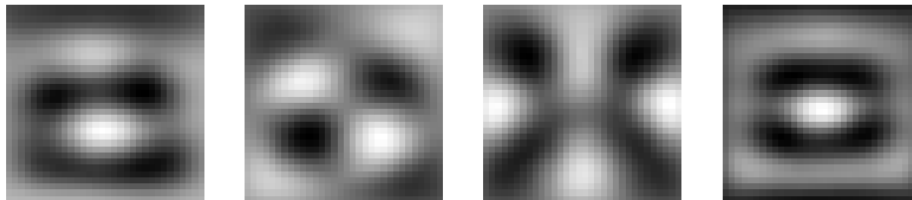
PCA top components (eigenvectors) 5 PCA top components (eigenvectors) 6 PCA top components (eigenvectors) 7 PCA top components (eigenvectors) 8



PCA top components (eigenvectors) 9 PCA top components (eigenvectors) 10 PCA top components (eigenvectors) 11 PCA top components (eigenvectors) 12



PCA top components (eigenvectors) 13 PCA top components (eigenvectors) 14 PCA top components (eigenvectors) 15 PCA top components (eigenvectors) 16



Building linear autoencoder with input_dim=1024, latent K=160

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1765098221.534824 1536 gpu_device.cc:2020] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 5580 MB memory: -> device: 0, name: NVIDIA GeForce RTX 3070 Laptop GPU, pci bus id: 0000:01:00.0, compute capability: 8.6

Start training linear autoencoder...

2025-12-07 09:03:44.143012: I tensorflow/core/framework/local_rendezvous.cc:407] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

Epoch 1/50 - train MSE: 0.227718

2025-12-07 09:03:44.444939: I tensorflow/core/framework/local_rendezvous.cc:407] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

```
2025-12-07 09:03:45.033278: I tensorflow/core/framework/local_rendezvous.cc:407]
Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

Epoch 5/50 - train MSE: 0.058654

2025-12-07 09:03:46.222011: I tensorflow/core/framework/local_rendezvous.cc:407]
Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

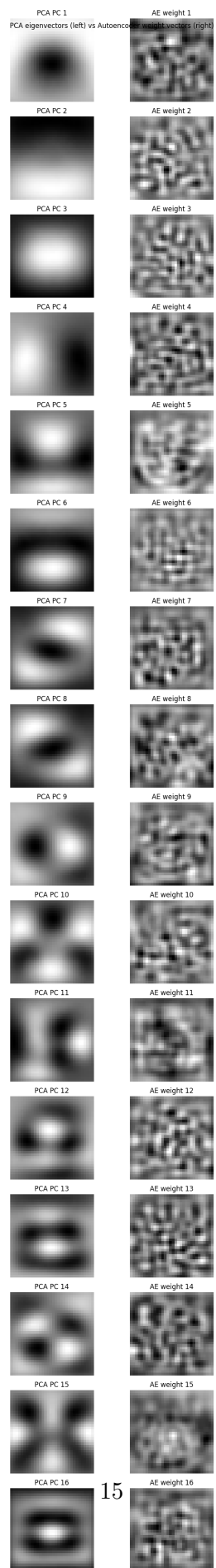
Epoch 10/50 - train MSE: 0.052516
Epoch 15/50 - train MSE: 0.052261

2025-12-07 09:03:48.590357: I tensorflow/core/framework/local_rendezvous.cc:407]
Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

Epoch 20/50 - train MSE: 0.052292
Epoch 25/50 - train MSE: 0.052236
Epoch 30/50 - train MSE: 0.052320

2025-12-07 09:03:53.543465: I tensorflow/core/framework/local_rendezvous.cc:407]
Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

Epoch 35/50 - train MSE: 0.052368
Epoch 40/50 - train MSE: 0.052351
Epoch 45/50 - train MSE: 0.052295
Epoch 50/50 - train MSE: 0.052396
Training finished.
Autoencoder reconstruction MSE on test set: 0.052593
W_components shape (for display): (160, 1024)
```



```

Cosine similarity PCA_1 vs AE_1: 0.0000
Cosine similarity PCA_2 vs AE_2: 0.0004
Cosine similarity PCA_3 vs AE_3: -0.0006
Cosine similarity PCA_4 vs AE_4: -0.0010
Cosine similarity PCA_5 vs AE_5: 0.1370
Cosine similarity PCA_6 vs AE_6: 0.0024
Cosine similarity PCA_7 vs AE_7: 0.0087
Cosine similarity PCA_8 vs AE_8: 0.0611
Cosine similarity PCA_9 vs AE_9: -0.1397
Cosine similarity PCA_10 vs AE_10: 0.0213
Cosine similarity PCA_11 vs AE_11: -0.2401
Cosine similarity PCA_12 vs AE_12: 0.0094
Cosine similarity PCA_13 vs AE_13: 0.0040
Cosine similarity PCA_14 vs AE_14: 0.0150
Cosine similarity PCA_15 vs AE_15: -0.2334
Cosine similarity PCA_16 vs AE_16: -0.0258
Average cosine similarity (top shown): -0.02383666424018608

```

Task 2 comments on weights vs PCA:

- The PCA components are smooth, global patterns since they are eigen based and are constrained to be orthogonal capturing maximum variance
- The AE weights are much noisier and more local since they are not having the same constraint of orthogonality as PCA and the objective is to reduce the error based on the architecture
- In AE it also depends on the activation function and number of layers since the weights may learn to match some localized features
- The feature map from PCA is more interpretable while that of AE may or may not be, regardless that does not mean the weight of AE may be bad since the learning method of AE is fundamentally different from that of PCA

0.0.3 Task 3: Deep Conv Autoencoder, Single-hidden-layer AE (sigmoid encoder, linear decoder) and 3-hidden-layer AE (sigmoid activations, linear decoder final).

```

[4]: X_train_img = X_train_scaled.reshape(-1, h, w, 1)
     X_test_img  = X_test_scaled.reshape(-1, h, w, 1)

print("Train / Test shapes (flat):", X_train_scaled.shape, X_test_scaled.shape)
print("Train / Test shapes (img):", X_train_img.shape, X_test_img.shape)

# PCA from Task 1
K = results_full['K']

# Deep Convolutional Autoencoder
def build_conv_autoencoder(input_shape=(h,w,1), latent_dim=K):

```



```

inp = keras.Input(shape=input_shape)
x = layers.Conv2D(32, (3,3), strides=1, padding='same',
↳activation='relu')(inp) # 32x32x32
x = layers.MaxPool2D((2,2), padding='same')(x) # 16x16x32
x = layers.Conv2D(64, (3,3), strides=1, padding='same',
↳activation='relu')(x) # 16x16x64
x = layers.MaxPool2D((2,2), padding='same')(x) # 8x8x64
x = layers.Conv2D(128, (3,3), strides=1, padding='same',
↳activation='relu')(x) # 8x8x128
x = layers.MaxPool2D((2,2), padding='same')(x) # 4x4x128
x = layers.Flatten()(x)
z = layers.Dense(latent_dim, name='latent')(x) # linear latent
# Decoder
x = layers.Dense(4*4*128, activation='relu')(z)
x = layers.Reshape((4,4,128))(x)
x = layers.UpSampling2D((2,2))(x) # 8x8x128
x = layers.Conv2D(128, (3,3), padding='same', activation='relu')(x)
x = layers.UpSampling2D((2,2))(x) # 16x16x128
x = layers.Conv2D(64, (3,3), padding='same', activation='relu')(x)
x = layers.UpSampling2D((2,2))(x) # 32x32x64
x = layers.Conv2D(32, (3,3), padding='same', activation='relu')(x)
# Final conv to single-channel linear output
x = layers.Conv2D(1, (3,3), padding='same', activation=None)(x) # linear
model = keras.Model(inputs=inp, outputs=x, name='conv_autoencoder')
return model

conv_ae = build_conv_autoencoder(input_shape=(h,w,1), latent_dim=K)
conv_ae.summary()

# Compile
conv_ae.compile(optimizer=keras.optimizers.Adam(1e-3), loss='mse')

# Train conv AE
epochs_conv = 30
batch_size = 256
history_conv = conv_ae.fit(
    X_train_img, X_train_img,
    validation_data=(X_test_img, X_test_img),
    epochs=epochs_conv, batch_size=batch_size, verbose=2
)

# Compute reconstruction on test set (flattened for MSE calculation)
X_test_recon_conv = conv_ae.predict(X_test_img, batch_size=256)
mse_conv = np.mean((X_test_img.reshape(len(X_test_img), -1) - X_test_recon_conv.
↳reshape(len(X_test_recon_conv), -1))**2)
print(f"[Conv AE] Test reconstruction MSE: {mse_conv:.6f}")

```

```

# Single-hidden-layer MLP Autoencoder with encoder sigmoid activation decoder
↳ linear activation
def build_single_hidden_mlp(input_dim=input_dim, latent_dim=K):
    inp = keras.Input(shape=(input_dim,))
    # Sigmoid encoder to K
    z = layers.Dense(latent_dim, activation='sigmoid', name='latent')(inp)
    # Linear decoder
    x_hat = layers.Dense(input_dim, activation=None)(z)
    model = keras.Model(inputs=inp, outputs=x_hat, name='single_hidden_mlp_ae')
    return model

mlp1 = build_single_hidden_mlp(input_dim=input_dim, latent_dim=K)
mlp1.summary()
mlp1.compile(optimizer=keras.optimizers.Adam(1e-3), loss='mse')

# Train on flattened standardized data
epochs_mlp = 50
history_mlp1 = mlp1.fit(
    X_train_scaled, X_train_scaled,
    validation_data=(X_test_scaled, X_test_scaled),
    epochs=epochs_mlp, batch_size=256, verbose=2
)

# Evaluate MSE on test set
X_test_recon_mlp1 = mlp1.predict(X_test_scaled, batch_size=256)
mse_mlp1 = np.mean((X_test_scaled - X_test_recon_mlp1)**2)
print(f"[Single-hidden MLP AE] Test reconstruction MSE: {mse_mlp1:.6f}")

# 3-hidden-layer MLP Autoencoder
# Implementation: encoder: Dense(h1, act='sigmoid') -> Dense(h2,
↳ act='sigmoid') -> Dense(latent=K, linear)
# Decoder mirrors encoder and final layer activation is linear.

import math
h_each = int(math.ceil(K / 3.0))
h1 = h2 = h_each
print("3-hidden-layer MLP AE config: h1 = h2 = approx K/3 ->", h1, h2, "latent
↳ K =", K)

def build_three_hidden_mlp(input_dim=input_dim, h1=h1, h2=h2, latent_dim=K):
    inp = keras.Input(shape=(input_dim,))
    x = layers.Dense(h1, activation='sigmoid')(inp)
    x = layers.Dense(h2, activation='sigmoid')(x)
    z = layers.Dense(latent_dim, activation='sigmoid', name='latent')(x) # use
↳ sigmoid at autoencoder
    x = layers.Dense(h2, activation='sigmoid')(z)
    x = layers.Dense(h1, activation='sigmoid')(x)

```

```

    x_hat = layers.Dense(input_dim, activation=None)(x) # final linear decoder
    model = keras.Model(inputs=inp, outputs=x_hat, name='three_hidden_mlp_ae')
    return model

mlp3 = build_three_hidden_mlp(input_dim=input_dim, h1=h1, h2=h2, latent_dim=K)
mlp3.summary()
mlp3.compile(optimizer=keras.optimizers.Adam(1e-3), loss='mse')

# Train
epochs_mlp3 = 50
history_mlp3 = mlp3.fit(
    X_train_scaled, X_train_scaled,
    validation_data=(X_test_scaled, X_test_scaled),
    epochs=epochs_mlp3, batch_size=256, verbose=2
)

# Evaluate
X_test_recon_mlp3 = mlp3.predict(X_test_scaled, batch_size=256)
mse_mlp3 = np.mean((X_test_scaled - X_test_recon_mlp3)**2)
print(f"[3-hidden-layer MLP AE] Test reconstruction MSE: {mse_mlp3:.6f}")

# Results
print("\n=== Reconstruction MSE (test set) ===")
print(f"Conv AE (deep conv) : {mse_conv:.6f}")
print(f"Single-hidden MLP (sigmoid->linear): {mse_mlp1:.6f}")
print(f"3-hidden-layer MLP (sigmoid, final linear): {mse_mlp3:.6f}")

# original vs reconstructions (grayscale)
def show_reconstructions(original_flat, recon_flat, n=8, title_prefix=""):
    idx = np.random.choice(len(original_flat), size=n, replace=False)
    fig, axes = plt.subplots(2, n, figsize=(n*1.6, 3.4))
    for i, j in enumerate(idx):
        orig = original_flat[j].reshape(h, w)
        rec = recon_flat[j].reshape(h, w)
        # un-standardize for display
        axes[0, i].imshow((orig - orig.min()) / (orig.max() - orig.min() + 1e-12), cmap='gray')
        axes[0, i].axis('off')
        axes[0, i].set_title("orig")
        axes[1, i].imshow((rec - rec.min()) / (rec.max() - rec.min() + 1e-12), cmap='gray')
        axes[1, i].axis('off')
        axes[1, i].set_title("recon")
    plt.suptitle(title_prefix)
    plt.show()

```

```
# For conv ae, use image-shaped outputs; convert to flattened comparable format
↳(same scaling)
show_reconstructions(X_test_scaled, X_test_recon_conv.
↳reshape(len(X_test_recon_conv), -1), n=6, title_prefix="Conv AE: original
↳(top) vs recon (bottom)")
show_reconstructions(X_test_scaled, X_test_recon_mlp1, n=6,
↳title_prefix="Single-hidden MLP AE: original vs recon")
show_reconstructions(X_test_scaled, X_test_recon_mlp3, n=6,
↳title_prefix="3-hidden-layer MLP AE: original vs recon")
```

Train / Test shapes (flat): (42000, 1024) (18000, 1024)

Train / Test shapes (img): (42000, 32, 32, 1) (18000, 32, 32, 1)

Model: "conv_autoencoder"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 32, 32, 1)	0
conv2d (Conv2D)	(None, 32, 32, 32)	320
max_pooling2d (MaxPooling2D)	(None, 16, 16, 32)	0
conv2d_1 (Conv2D)	(None, 16, 16, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 64)	0
conv2d_2 (Conv2D)	(None, 8, 8, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 128)	0
flatten (Flatten)	(None, 2048)	0
latent (Dense)	(None, 160)	327,840
dense (Dense)	(None, 2048)	329,728
reshape (Reshape)	(None, 4, 4, 128)	0
up_sampling2d (UpSampling2D)	(None, 8, 8, 128)	0
conv2d_3 (Conv2D)	(None, 8, 8, 128)	147,584
up_sampling2d_1 (UpSampling2D)	(None, 16, 16, 128)	0
conv2d_4 (Conv2D)	(None, 16, 16, 64)	73,792

up_sampling2d_2 (UpSampling2D)	(None, 32, 32, 64)	0
conv2d_5 (Conv2D)	(None, 32, 32, 32)	18,464
conv2d_6 (Conv2D)	(None, 32, 32, 1)	289

Total params: 990,369 (3.78 MB)

Trainable params: 990,369 (3.78 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/30

2025-12-07 09:04:02.074799: I external/local_xla/xla/service/service.cc:163] XLA service 0x712948a2f2a0 initialized for platform CUDA (this does not guarantee that XLA will be used). Devices:

2025-12-07 09:04:02.074824: I external/local_xla/xla/service/service.cc:171] StreamExecutor device (0): NVIDIA GeForce RTX 3070 Laptop GPU, Compute Capability 8.6

2025-12-07 09:04:02.126827: I tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_util.cc:269] disabling MLIR crash reproducer, set env var `MLIR\_CRASH\_REPRODUCER\_DIRECTORY` to enable.

2025-12-07 09:04:02.397748: I external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:473] Loaded cuDNN version 91700

2025-12-07 09:04:03.412342: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function 'gemm_fusion_dot_1052', 68 bytes spill stores, 68 bytes spill loads

2025-12-07 09:04:03.419835: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function 'gemm_fusion_dot_1054', 4 bytes spill stores, 4 bytes spill loads

2025-12-07 09:04:03.450719: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function 'gemm_fusion_dot_1054', 144 bytes spill stores, 144 bytes spill loads

2025-12-07 09:04:03.596124: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function

'gemm_fusion_dot_912', 11580 bytes spill stores, 10236 bytes spill loads

2025-12-07 09:04:03.675115: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_383', 11088 bytes spill stores, 11284 bytes spill loads

I0000 00:00:1765098249.442036 4610 device_compiler.h:196] Compiled cluster using XLA! This line is logged at most once for the lifetime of the process.

165/165 - 18s - 108ms/step - loss: 0.3755 - val_loss: 0.2617

Epoch 2/30

165/165 - 4s - 23ms/step - loss: 0.2050 - val_loss: 0.2075

Epoch 3/30

165/165 - 4s - 22ms/step - loss: 0.1718 - val_loss: 0.1676

Epoch 4/30

165/165 - 5s - 28ms/step - loss: 0.1536 - val_loss: 0.1730

Epoch 5/30

165/165 - 4s - 22ms/step - loss: 0.1412 - val_loss: 0.1398

Epoch 6/30

165/165 - 4s - 22ms/step - loss: 0.1306 - val_loss: 0.1311

Epoch 7/30

165/165 - 4s - 22ms/step - loss: 0.1226 - val_loss: 0.1237

Epoch 8/30

165/165 - 4s - 22ms/step - loss: 0.1167 - val_loss: 0.1210

Epoch 9/30

165/165 - 4s - 21ms/step - loss: 0.1116 - val_loss: 0.1144

Epoch 10/30

165/165 - 4s - 21ms/step - loss: 0.1052 - val_loss: 0.1142

Epoch 11/30

165/165 - 4s - 21ms/step - loss: 0.1020 - val_loss: 0.1145

Epoch 12/30

165/165 - 3s - 17ms/step - loss: 0.0994 - val_loss: 0.1130

Epoch 13/30

165/165 - 4s - 21ms/step - loss: 0.0946 - val_loss: 0.0955

Epoch 14/30

165/165 - 4s - 22ms/step - loss: 0.0902 - val_loss: 0.0995

Epoch 15/30

165/165 - 4s - 23ms/step - loss: 0.0892 - val_loss: 0.0981

Epoch 16/30

165/165 - 4s - 23ms/step - loss: 0.0855 - val_loss: 0.0919

Epoch 17/30

165/165 - 4s - 22ms/step - loss: 0.0833 - val_loss: 0.0946

Epoch 18/30

165/165 - 4s - 22ms/step - loss: 0.0853 - val_loss: 0.1000

Epoch 19/30

165/165 - 4s - 22ms/step - loss: 0.0800 - val_loss: 0.0921

Epoch 20/30

```

165/165 - 4s - 22ms/step - loss: 0.0769 - val_loss: 0.1038
Epoch 21/30
165/165 - 3s - 18ms/step - loss: 0.0819 - val_loss: 0.0850
Epoch 22/30
165/165 - 3s - 21ms/step - loss: 0.0740 - val_loss: 0.0786
Epoch 23/30
165/165 - 4s - 22ms/step - loss: 0.0726 - val_loss: 0.0858
Epoch 24/30
165/165 - 4s - 22ms/step - loss: 0.0734 - val_loss: 0.0831
Epoch 25/30
165/165 - 4s - 22ms/step - loss: 0.0729 - val_loss: 0.0844
Epoch 26/30
165/165 - 4s - 22ms/step - loss: 0.0711 - val_loss: 0.0777
Epoch 27/30
165/165 - 4s - 22ms/step - loss: 0.0690 - val_loss: 0.0779
Epoch 28/30
165/165 - 4s - 22ms/step - loss: 0.0691 - val_loss: 0.0843
Epoch 29/30
165/165 - 4s - 22ms/step - loss: 0.0698 - val_loss: 0.0766
Epoch 30/30
165/165 - 3s - 18ms/step - loss: 0.0684 - val_loss: 0.0777
71/71          2s 15ms/step
[Conv AE] Test reconstruction MSE: 0.077688
Model: "single_hidden_mlp_ae"

```

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None , 1024)	0
latent (Dense)	(None , 160)	164,000
dense_1 (Dense)	(None , 1024)	164,864

Total params: [328,864](#) (1.25 MB)

Trainable params: [328,864](#) (1.25 MB)

Non-trainable params: [0](#) (0.00 B)

Epoch 1/50

2025-12-07 09:06:05.529604: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function
'gemm_fusion_dot_252', 636 bytes spill stores, 636 bytes spill loads

2025-12-07 09:06:05.691157: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_123', 10152 bytes spill stores, 10404 bytes spill loads

165/165 - 4s - 24ms/step - loss: 0.4330 - val_loss: 0.2758
Epoch 2/50
165/165 - 1s - 5ms/step - loss: 0.2310 - val_loss: 0.2024
Epoch 3/50
165/165 - 1s - 5ms/step - loss: 0.1790 - val_loss: 0.1660
Epoch 4/50
165/165 - 1s - 3ms/step - loss: 0.1511 - val_loss: 0.1440
Epoch 5/50
165/165 - 1s - 3ms/step - loss: 0.1329 - val_loss: 0.1289
Epoch 6/50
165/165 - 1s - 3ms/step - loss: 0.1198 - val_loss: 0.1183
Epoch 7/50
165/165 - 1s - 4ms/step - loss: 0.1101 - val_loss: 0.1092
Epoch 8/50
165/165 - 1s - 3ms/step - loss: 0.1024 - val_loss: 0.1029
Epoch 9/50
165/165 - 1s - 4ms/step - loss: 0.0963 - val_loss: 0.0973
Epoch 10/50
165/165 - 1s - 3ms/step - loss: 0.0912 - val_loss: 0.0925
Epoch 11/50
165/165 - 1s - 4ms/step - loss: 0.0871 - val_loss: 0.0884
Epoch 12/50
165/165 - 1s - 4ms/step - loss: 0.0835 - val_loss: 0.0852
Epoch 13/50
165/165 - 1s - 3ms/step - loss: 0.0806 - val_loss: 0.0826
Epoch 14/50
165/165 - 1s - 3ms/step - loss: 0.0782 - val_loss: 0.0805
Epoch 15/50
165/165 - 1s - 4ms/step - loss: 0.0761 - val_loss: 0.0787
Epoch 16/50
165/165 - 1s - 4ms/step - loss: 0.0743 - val_loss: 0.0771
Epoch 17/50
165/165 - 1s - 4ms/step - loss: 0.0728 - val_loss: 0.0754
Epoch 18/50
165/165 - 1s - 3ms/step - loss: 0.0714 - val_loss: 0.0740
Epoch 19/50
165/165 - 1s - 3ms/step - loss: 0.0704 - val_loss: 0.0726
Epoch 20/50
165/165 - 1s - 3ms/step - loss: 0.0692 - val_loss: 0.0718

Epoch 21/50
165/165 - 1s - 4ms/step - loss: 0.0686 - val_loss: 0.0708
Epoch 22/50
165/165 - 1s - 3ms/step - loss: 0.0675 - val_loss: 0.0701
Epoch 23/50
165/165 - 1s - 3ms/step - loss: 0.0670 - val_loss: 0.0694
Epoch 24/50
165/165 - 1s - 3ms/step - loss: 0.0663 - val_loss: 0.0687
Epoch 25/50
165/165 - 1s - 3ms/step - loss: 0.0655 - val_loss: 0.0681
Epoch 26/50
165/165 - 1s - 3ms/step - loss: 0.0651 - val_loss: 0.0678
Epoch 27/50
165/165 - 1s - 4ms/step - loss: 0.0647 - val_loss: 0.0674
Epoch 28/50
165/165 - 1s - 4ms/step - loss: 0.0642 - val_loss: 0.0668
Epoch 29/50
165/165 - 1s - 3ms/step - loss: 0.0637 - val_loss: 0.0664
Epoch 30/50
165/165 - 1s - 3ms/step - loss: 0.0633 - val_loss: 0.0659
Epoch 31/50
165/165 - 1s - 4ms/step - loss: 0.0636 - val_loss: 0.0654
Epoch 32/50
165/165 - 1s - 3ms/step - loss: 0.0625 - val_loss: 0.0649
Epoch 33/50
165/165 - 1s - 4ms/step - loss: 0.0623 - val_loss: 0.0647
Epoch 34/50
165/165 - 1s - 4ms/step - loss: 0.0621 - val_loss: 0.0644
Epoch 35/50
165/165 - 1s - 3ms/step - loss: 0.0618 - val_loss: 0.0644
Epoch 36/50
165/165 - 1s - 4ms/step - loss: 0.0616 - val_loss: 0.0647
Epoch 37/50
165/165 - 1s - 3ms/step - loss: 0.0613 - val_loss: 0.0636
Epoch 38/50
165/165 - 1s - 4ms/step - loss: 0.0610 - val_loss: 0.0635
Epoch 39/50
165/165 - 1s - 3ms/step - loss: 0.0611 - val_loss: 0.0626
Epoch 40/50
165/165 - 1s - 3ms/step - loss: 0.0603 - val_loss: 0.0623
Epoch 41/50
165/165 - 1s - 3ms/step - loss: 0.0602 - val_loss: 0.0622
Epoch 42/50
165/165 - 1s - 3ms/step - loss: 0.0602 - val_loss: 0.0618
Epoch 43/50
165/165 - 1s - 3ms/step - loss: 0.0599 - val_loss: 0.0616
Epoch 44/50
165/165 - 1s - 3ms/step - loss: 0.0598 - val_loss: 0.0619

Epoch 45/50
 165/165 - 0s - 441us/step - loss: 0.0597 - val_loss: 0.0620
 Epoch 46/50
 165/165 - 1s - 3ms/step - loss: 0.0596 - val_loss: 0.0619
 Epoch 47/50
 165/165 - 1s - 3ms/step - loss: 0.0595 - val_loss: 0.0616
 Epoch 48/50
 165/165 - 1s - 3ms/step - loss: 0.0593 - val_loss: 0.0613
 Epoch 49/50
 165/165 - 1s - 3ms/step - loss: 0.0590 - val_loss: 0.0611
 Epoch 50/50
 165/165 - 1s - 4ms/step - loss: 0.0589 - val_loss: 0.0610
 71/71 0s 4ms/step
 [Single-hidden MLP AE] Test reconstruction MSE: 0.060972
 3-hidden-layer MLP AE config: h1 = h2 = approx K/3 -> 54 54 latent K = 160
 Model: "three_hidden_mlp_ae"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None , 1024)	0
dense_2 (Dense)	(None , 54)	55,350
dense_3 (Dense)	(None , 54)	2,970
latent (Dense)	(None , 160)	8,800
dense_4 (Dense)	(None , 54)	8,694
dense_5 (Dense)	(None , 54)	2,970
dense_6 (Dense)	(None , 1024)	56,320

Total params: [135,104](#) (527.75 KB)

Trainable params: [135,104](#) (527.75 KB)

Non-trainable params: [0](#) (0.00 B)

Epoch 1/50

2025-12-07 09:06:39.084258: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All

configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:39.084322: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
 configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:39.084341: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
 configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:39.084405: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
 configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:39.084418: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
 configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:39.084431: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
 configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:39.084467: I
 external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
 configs were filtered out because none of them sufficiently match the hints.
 Maybe the hints set does not contain a good representative set of valid configs?
 Working around this by using the full hints set instead.

2025-12-07 09:06:40.501892: I
 external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
 warning : Registers are spilled to local memory in function
 'gemm_fusion_dot_264', 220 bytes spill stores, 220 bytes spill loads

2025-12-07 09:06:41.159346: I
 external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
 warning : Registers are spilled to local memory in function
 'gemm_fusion_dot_269', 72 bytes spill stores, 72 bytes spill loads

2025-12-07 09:06:42.044078: I
 external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
 warning : Registers are spilled to local memory in function
 'gemm_fusion_dot_475', 180 bytes spill stores, 180 bytes spill loads

```

2025-12-07 09:06:42.288929: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_421', 140 bytes spill stores, 140 bytes spill loads

2025-12-07 09:06:42.390312: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_434', 92 bytes spill stores, 92 bytes spill loads

2025-12-07 09:06:42.471521: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_475', 4 bytes spill stores, 4 bytes spill loads

2025-12-07 09:06:42.607180: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_475', 12 bytes spill stores, 12 bytes spill loads

2025-12-07 09:06:43.139690: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_475', 1176 bytes spill stores, 1176 bytes spill loads

2025-12-07 09:06:48.900366: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.
2025-12-07 09:06:48.900417: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.
2025-12-07 09:06:48.900432: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.
2025-12-07 09:06:48.900456: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.
2025-12-07 09:06:49.651850: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function

```

'gemm_fusion_dot_31', 20 bytes spill stores, 20 bytes spill loads

2025-12-07 09:06:50.129739: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_36', 4 bytes spill stores, 4 bytes spill loads

2025-12-07 09:06:50.232589: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_31', 32 bytes spill stores, 32 bytes spill loads

2025-12-07 09:06:50.438032: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_26', 16 bytes spill stores, 16 bytes spill loads

2025-12-07 09:06:50.490903: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_46', 16 bytes spill stores, 16 bytes spill loads

2025-12-07 09:06:50.662142: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_46', 8 bytes spill stores, 8 bytes spill loads

2025-12-07 09:06:50.719948: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_36', 12 bytes spill stores, 12 bytes spill loads

2025-12-07 09:06:50.792223: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_46', 184 bytes spill stores, 184 bytes spill loads

2025-12-07 09:06:50.898609: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas

warning : Registers are spilled to local memory in function

'gemm_fusion_dot_31', 24 bytes spill stores, 24 bytes spill loads

165/165 - 14s - 86ms/step - loss: 0.8874 - val_loss: 0.7429

Epoch 2/50

165/165 - 1s - 4ms/step - loss: 0.7229 - val_loss: 0.7151

Epoch 3/50

165/165 - 1s - 4ms/step - loss: 0.7100 - val_loss: 0.7245

Epoch 4/50
165/165 - 1s - 4ms/step - loss: 0.6559 - val_loss: 0.6122
Epoch 5/50
165/165 - 1s - 4ms/step - loss: 0.5988 - val_loss: 0.6067
Epoch 6/50
165/165 - 1s - 3ms/step - loss: 0.5903 - val_loss: 0.5900
Epoch 7/50
165/165 - 1s - 4ms/step - loss: 0.5870 - val_loss: 0.5866
Epoch 8/50
165/165 - 1s - 4ms/step - loss: 0.5859 - val_loss: 0.5856
Epoch 9/50
165/165 - 1s - 7ms/step - loss: 0.5854 - val_loss: 0.5848
Epoch 10/50
165/165 - 1s - 4ms/step - loss: 0.5851 - val_loss: 0.5851
Epoch 11/50
165/165 - 1s - 3ms/step - loss: 0.5844 - val_loss: 0.5839
Epoch 12/50
165/165 - 1s - 3ms/step - loss: 0.5839 - val_loss: 0.5835
Epoch 13/50
165/165 - 1s - 3ms/step - loss: 0.5834 - val_loss: 0.5835
Epoch 14/50
165/165 - 1s - 5ms/step - loss: 0.5829 - val_loss: 0.5827
Epoch 15/50
165/165 - 1s - 3ms/step - loss: 0.5819 - val_loss: 0.5808
Epoch 16/50
165/165 - 1s - 3ms/step - loss: 0.5544 - val_loss: 0.5189
Epoch 17/50
165/165 - 1s - 4ms/step - loss: 0.5139 - val_loss: 0.5102
Epoch 18/50
165/165 - 1s - 5ms/step - loss: 0.5087 - val_loss: 0.5091
Epoch 19/50
165/165 - 1s - 3ms/step - loss: 0.5070 - val_loss: 0.5073
Epoch 20/50
165/165 - 1s - 3ms/step - loss: 0.5062 - val_loss: 0.5060
Epoch 21/50
165/165 - 1s - 4ms/step - loss: 0.5058 - val_loss: 0.5058
Epoch 22/50
165/165 - 1s - 4ms/step - loss: 0.5050 - val_loss: 0.5043
Epoch 23/50
165/165 - -0s - -1148us/step - loss: 0.5045 - val_loss: 0.5038
Epoch 24/50
165/165 - 1s - 3ms/step - loss: 0.5040 - val_loss: 0.5033
Epoch 25/50
165/165 - 1s - 4ms/step - loss: 0.5036 - val_loss: 0.5031
Epoch 26/50
165/165 - 1s - 4ms/step - loss: 0.5032 - val_loss: 0.5035
Epoch 27/50
165/165 - 1s - 4ms/step - loss: 0.5025 - val_loss: 0.5016

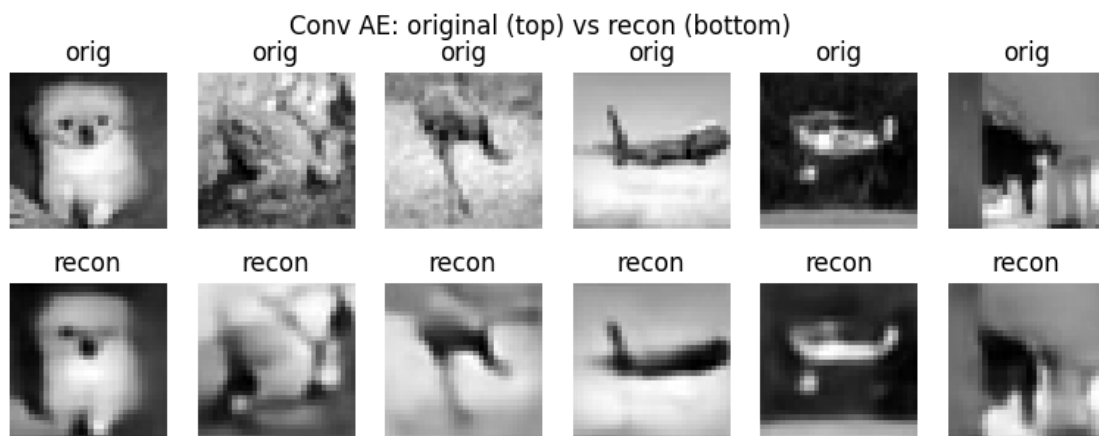
Epoch 28/50
 165/165 - 1s - 3ms/step - loss: 0.4994 - val_loss: 0.4921
 Epoch 29/50
 165/165 - 1s - 4ms/step - loss: 0.4825 - val_loss: 0.4788
 Epoch 30/50
 165/165 - 1s - 3ms/step - loss: 0.4769 - val_loss: 0.4773
 Epoch 31/50
 165/165 - 1s - 4ms/step - loss: 0.4746 - val_loss: 0.4743
 Epoch 32/50
 165/165 - 1s - 4ms/step - loss: 0.4700 - val_loss: 0.4607
 Epoch 33/50
 165/165 - 1s - 7ms/step - loss: 0.4510 - val_loss: 0.4500
 Epoch 34/50
 165/165 - 1s - 4ms/step - loss: 0.4441 - val_loss: 0.4426
 Epoch 35/50
 165/165 - 1s - 4ms/step - loss: 0.4405 - val_loss: 0.4396
 Epoch 36/50
 165/165 - 1s - 4ms/step - loss: 0.4379 - val_loss: 0.4384
 Epoch 37/50
 165/165 - 1s - 4ms/step - loss: 0.4357 - val_loss: 0.4364
 Epoch 38/50
 165/165 - 1s - 4ms/step - loss: 0.4342 - val_loss: 0.4349
 Epoch 39/50
 165/165 - 1s - 4ms/step - loss: 0.4331 - val_loss: 0.4336
 Epoch 40/50
 165/165 - 1s - 4ms/step - loss: 0.4323 - val_loss: 0.4331
 Epoch 41/50
 165/165 - 1s - 4ms/step - loss: 0.4317 - val_loss: 0.4326
 Epoch 42/50
 165/165 - 1s - 3ms/step - loss: 0.4313 - val_loss: 0.4324
 Epoch 43/50
 165/165 - 1s - 3ms/step - loss: 0.4310 - val_loss: 0.4328
 Epoch 44/50
 165/165 - 1s - 3ms/step - loss: 0.4311 - val_loss: 0.4320
 Epoch 45/50
 165/165 - 1s - 4ms/step - loss: 0.4306 - val_loss: 0.4323
 Epoch 46/50
 165/165 - 1s - 3ms/step - loss: 0.4302 - val_loss: 0.4318
 Epoch 47/50
 165/165 - 1s - 3ms/step - loss: 0.4297 - val_loss: 0.4309
 Epoch 48/50
 165/165 - 1s - 4ms/step - loss: 0.4293 - val_loss: 0.4308
 Epoch 49/50
 165/165 - 1s - 4ms/step - loss: 0.4291 - val_loss: 0.4306
 Epoch 50/50
 165/165 - 1s - 4ms/step - loss: 0.4290 - val_loss: 0.4303
 71/71 1s 11ms/step
 [3-hidden-layer MLP AE] Test reconstruction MSE: 0.430340

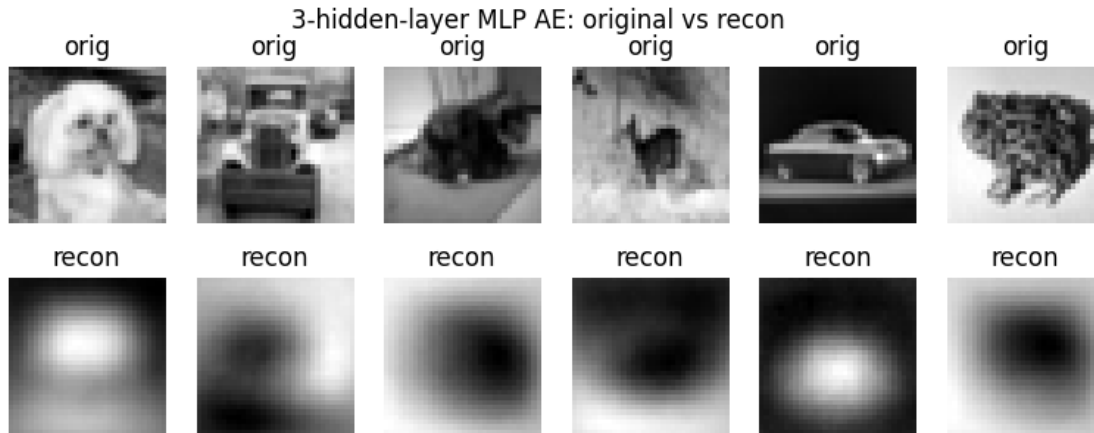
=== Reconstruction MSE (test set) ===

Conv AE (deep conv) : 0.077688

Single-hidden MLP (sigmoid->linear): 0.060972

3-hidden-layer MLP (sigmoid, final linear): 0.430340





0.0.4 Task 4: MNIST -> Conv Autoencoder -> extract features -> 7-output MLP (7-seg LED) -> confusion matrix

```
[5]: # Load MNIST and preprocess
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()
# Normalize to [0,1]
x_train = x_train.astype('float32') / 255.0
x_test  = x_test.astype('float32') / 255.0

# reshape
x_train = x_train[..., np.newaxis]
x_test  = x_test[..., np.newaxis]

print("MNIST shapes:", x_train.shape, x_test.shape, "Labels:", np.
      ↪unique(y_train))

# 7-segment mapping egment order: top, top-right, bottom-right, d = bottom, e_
      ↪bottom-left, f = top-left, g = middle
seg_map = {
    0: [1,1,1,1,1,1,0],
    1: [0,1,1,0,0,0,0],
    2: [1,1,0,1,1,0,1],
    3: [1,1,1,1,0,0,1],
    4: [0,1,1,0,0,1,1],
    5: [1,0,1,1,0,1,1],
    6: [1,0,1,1,1,1,1],
    7: [1,1,1,0,0,0,0],
    8: [1,1,1,1,1,1,1],
    9: [1,1,1,1,0,1,1],
}
# Precompute arrays
```

```

seg_map_arr = {d: np.array(bits, dtype=np.int32) for d, bits in seg_map.items()}
# For quick reverse lookup: map tuple(bits) -> digit
seg_tuple_to_digit = {tuple(v.tolist()): k for k, v in seg_map_arr.items()}

# Build ground-truth 7-bit labels for MNIST digits
y_train_seg = np.array([seg_map_arr[int(d)] for d in y_train])
y_test_seg = np.array([seg_map_arr[int(d)] for d in y_test])

# Build a small conv autoencoder and train (encoder -> latent features)
input_shape = x_train.shape[1:]
latent_dim = 64 # set latent dimension

def build_conv_autoencoder(input_shape=(28,28,1), latent_dim=64):
    inp = keras.Input(shape=input_shape)
    x = layers.Conv2D(32, 3, padding='same', activation='relu')(inp)
    x = layers.MaxPool2D(2, padding='same')(x)
    x = layers.Conv2D(64, 3, padding='same', activation='relu')(x)
    x = layers.MaxPool2D(2, padding='same')(x)
    x = layers.Flatten()(x)
    z = layers.Dense(latent_dim, name='latent')(x) # linear latent
    # decoder
    x = layers.Dense(7*7*64, activation='relu')(z)
    x = layers.Reshape((7,7,64))(x)
    x = layers.UpSampling2D(2)(x)
    x = layers.Conv2D(64, 3, padding='same', activation='relu')(x)
    x = layers.UpSampling2D(2)(x)
    x = layers.Conv2D(32, 3, padding='same', activation='relu')(x)
    out = layers.Conv2D(1, 3, padding='same', activation=None)(x) # linear
    ↪ output
    model = keras.Model(inputs=inp, outputs=out, name='conv_autoencoder')
    # encoder model separately
    encoder = keras.Model(inputs=inp, outputs=z, name='encoder')
    return model, encoder

conv_ae, encoder = build_conv_autoencoder(input_shape=input_shape,
    ↪ latent_dim=latent_dim)
conv_ae.compile(optimizer=keras.optimizers.Adam(1e-3), loss='mse')
conv_ae.summary()

# Train with early stopping to save time
callbacks = [
    keras.callbacks.EarlyStopping(monitor='val_loss', patience=3,
    ↪ restore_best_weights=True),
    keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.5,
    ↪ patience=2)
]
history = conv_ae.fit(

```

```

x_train, x_train,
validation_data=(x_test, x_test),
epochs=25,
batch_size=256,
callbacks=callbacks,
verbose=2
)

# show a few reconstructions
def show_recons(orig, recon, n=6, title="Reconstructions"):
    idx = np.random.choice(len(orig), size=n, replace=False)
    fig, axes = plt.subplots(2, n, figsize=(n*1.5, 3))
    for i,k in enumerate(idx):
        axes[0,i].imshow(orig[k].squeeze(), cmap='gray'); axes[0,i].axis('off');
        ↪ axes[0,i].set_title("orig")
        axes[1,i].imshow(recon[k].squeeze(), cmap='gray'); axes[1,i].
        ↪ axis('off'); axes[1,i].set_title("recon")
    plt.suptitle(title); plt.show()

x_test_recon = conv_ae.predict(x_test, batch_size=512)
show_recons(x_test, x_test_recon, n=6, title="Conv AE: original vs_
↪ reconstruction (MNIST)")

# Extract latent features for train & test
Z_train = encoder.predict(x_train, batch_size=512)
Z_test = encoder.predict(x_test, batch_size=512)
print("Latent features shapes:", Z_train.shape, Z_test.shape)

# Build MLP classifier with 7 outputs (sigmoid)
def build_7seg_mlp(input_dim, hidden=128):
    inp = keras.Input(shape=(input_dim,))
    x = layers.Dense(hidden, activation='relu')(inp)
    x = layers.Dropout(0.2)(x)
    x = layers.Dense(hidden//2, activation='relu')(x)
    out = layers.Dense(7, activation='sigmoid')(x) # 7 outputs, sigmoid for_
    ↪ multi-label
    model = keras.Model(inputs=inp, outputs=out, name='7seg_mlp')
    return model

mlp_7seg = build_7seg_mlp(Z_train.shape[1], hidden=128)
mlp_7seg.compile(optimizer=keras.optimizers.Adam(1e-3),_
    ↪ loss='binary_crossentropy', metrics=['accuracy'])
mlp_7seg.summary()

# Train the MLP on latent features to predict the 7-bit vectors
history_mlp = mlp_7seg.fit(
    Z_train, y_train_seg,

```

```

validation_data=(Z_test, y_test_seg),
epochs=30,
batch_size=256,
callbacks=callbacks,
verbose=2
)

# Inference: predict 7 segments on test set, then map to digit
y_prob_seg = mlp_7seg.predict(Z_test, batch_size=512)
y_pred_seg_bin = (y_prob_seg >= 0.5).astype(int)

# Map 7-bit pattern to digit:
def seg_bits_to_digit(bits_row):
    tup = tuple(int(b) for b in bits_row.tolist())
    if tup in seg_tuple_to_digit:
        return seg_tuple_to_digit[tup]
    # else pick nearest by Hamming distance:
    best_d, best_hd = None, 999
    for d, patt in seg_map_arr.items():
        hd = np.sum(np.abs(patt - bits_row))
        if hd < best_hd:
            best_hd = hd; best_d = d
    return int(best_d)

# Apply mapping for all test samples
y_pred_digits = np.array([seg_bits_to_digit(row) for row in y_pred_seg_bin])
y_true_digits = y_test.copy()

# Also compute per-segment accuracy
segment_acc = (y_pred_seg_bin == y_test_seg).mean(axis=0)
for i, acc in enumerate(segment_acc):
    print(f"Segment {i} accuracy: {acc*100:.2f}%")
print("Average segment accuracy: {:.2f}%".format(segment_acc.mean()*100))

# Digit-level accuracy (after mapping)
digit_acc = accuracy_score(y_true_digits, y_pred_digits)
print("Digit accuracy (mapped from 7-seg predictions): {:.2f}%".
      format(digit_acc*100))

# Confusion matrix (digits) & plot
cm = confusion_matrix(y_true_digits, y_pred_digits, labels=np.arange(10))
plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=np.arange(10),
            yticklabels=np.arange(10))
plt.xlabel("Predicted digit (via 7-seg mapping)")
plt.ylabel("True digit")
plt.title("Confusion Matrix: True vs Predicted digits (via 7-seg classifier)")

```

```
plt.show()

# Also show a few examples where mapping failed
mismatch_idx = np.where(y_true_digits != y_pred_digits)[0]
print(f"Total mismatches: {len(mismatch_idx)} / {len(y_test)}")
for i in mismatch_idx[:8]:
    print("True:", y_true_digits[i], " Predicted:", y_pred_digits[i], " Pred_
↪seg:", y_pred_seg_bin[i], "Prob seg:", np.round(y_prob_seg[i],3))
    plt.figure(figsize=(2,2)); plt.imshow(x_test[i].squeeze(), cmap='gray');
↪plt.axis('off'); plt.title(f"True {y_true_digits[i]} Pred_
↪{y_pred_digits[i]}"); plt.show()
```

MNIST shapes: (60000, 28, 28, 1) (10000, 28, 28, 1) Labels: [0 1 2 3 4 5 6 7 8 9]

Model: "conv_autoencoder"

Layer (type)	Output Shape	Param #
input_layer_3 (InputLayer)	(None, 28, 28, 1)	0
conv2d_7 (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d_3 (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_8 (Conv2D)	(None, 14, 14, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 7, 7, 64)	0
flatten_1 (Flatten)	(None, 3136)	0
latent (Dense)	(None, 64)	200,768
dense_7 (Dense)	(None, 3136)	203,840
reshape_1 (Reshape)	(None, 7, 7, 64)	0
up_sampling2d_3 (UpSampling2D)	(None, 14, 14, 64)	0
conv2d_9 (Conv2D)	(None, 14, 14, 64)	36,928
up_sampling2d_4 (UpSampling2D)	(None, 28, 28, 64)	0
conv2d_10 (Conv2D)	(None, 28, 28, 32)	18,464
conv2d_11 (Conv2D)	(None, 28, 28, 1)	289

Total params: 479,105 (1.83 MB)

Trainable params: 479,105 (1.83 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/25

2025-12-07 09:07:34.596913: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_299', 8 bytes spill stores, 8 bytes spill loads

235/235 - 12s - 52ms/step - loss: 0.0209 - val_loss: 0.0073 - learning_rate:
0.0010

Epoch 2/25

235/235 - 3s - 12ms/step - loss: 0.0060 - val_loss: 0.0049 - learning_rate:
0.0010

Epoch 3/25

235/235 - 3s - 12ms/step - loss: 0.0045 - val_loss: 0.0040 - learning_rate:
0.0010

Epoch 4/25

235/235 - 3s - 12ms/step - loss: 0.0039 - val_loss: 0.0036 - learning_rate:
0.0010

Epoch 5/25

235/235 - 3s - 14ms/step - loss: 0.0035 - val_loss: 0.0033 - learning_rate:
0.0010

Epoch 6/25

235/235 - 3s - 13ms/step - loss: 0.0031 - val_loss: 0.0030 - learning_rate:
0.0010

Epoch 7/25

235/235 - 5s - 23ms/step - loss: 0.0029 - val_loss: 0.0029 - learning_rate:
0.0010

Epoch 8/25

235/235 - 3s - 13ms/step - loss: 0.0028 - val_loss: 0.0027 - learning_rate:
0.0010

Epoch 9/25

235/235 - 3s - 14ms/step - loss: 0.0026 - val_loss: 0.0026 - learning_rate:
0.0010

Epoch 10/25

235/235 - 3s - 14ms/step - loss: 0.0025 - val_loss: 0.0027 - learning_rate:
0.0010

Epoch 11/25

235/235 - 2s - 9ms/step - loss: 0.0024 - val_loss: 0.0024 - learning_rate:
0.0010

Epoch 12/25
235/235 - 3s - 13ms/step - loss: 0.0023 - val_loss: 0.0024 - learning_rate: 0.0010

Epoch 13/25
235/235 - 3s - 13ms/step - loss: 0.0022 - val_loss: 0.0023 - learning_rate: 0.0010

Epoch 14/25
235/235 - 3s - 13ms/step - loss: 0.0022 - val_loss: 0.0023 - learning_rate: 0.0010

Epoch 15/25
235/235 - 3s - 14ms/step - loss: 0.0021 - val_loss: 0.0022 - learning_rate: 0.0010

Epoch 16/25
235/235 - 3s - 13ms/step - loss: 0.0020 - val_loss: 0.0022 - learning_rate: 0.0010

Epoch 17/25
235/235 - 3s - 13ms/step - loss: 0.0020 - val_loss: 0.0022 - learning_rate: 0.0010

Epoch 18/25
235/235 - 3s - 13ms/step - loss: 0.0018 - val_loss: 0.0019 - learning_rate: 5.0000e-04

Epoch 19/25
235/235 - 3s - 13ms/step - loss: 0.0018 - val_loss: 0.0019 - learning_rate: 5.0000e-04

Epoch 20/25
235/235 - 3s - 13ms/step - loss: 0.0018 - val_loss: 0.0019 - learning_rate: 5.0000e-04

Epoch 21/25
235/235 - 3s - 13ms/step - loss: 0.0017 - val_loss: 0.0018 - learning_rate: 2.5000e-04

Epoch 22/25
235/235 - 2s - 10ms/step - loss: 0.0017 - val_loss: 0.0018 - learning_rate: 2.5000e-04

Epoch 23/25
235/235 - 3s - 13ms/step - loss: 0.0017 - val_loss: 0.0018 - learning_rate: 2.5000e-04

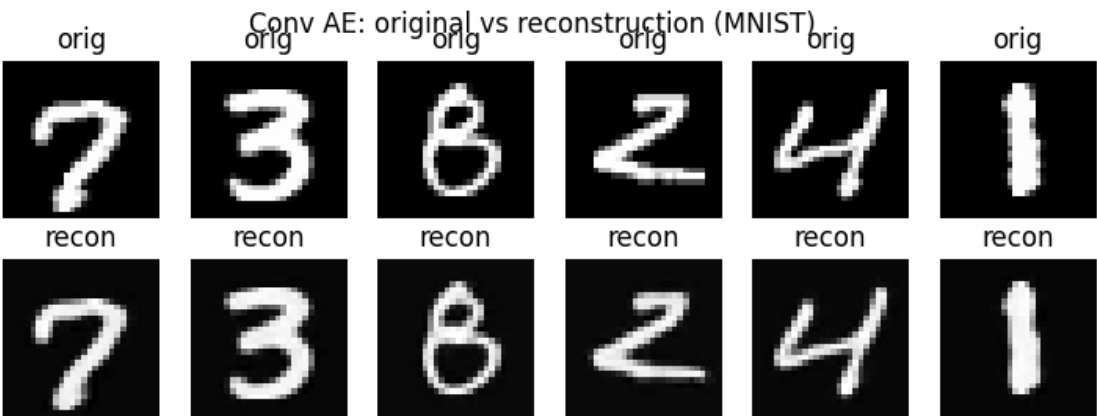
Epoch 24/25
235/235 - 3s - 13ms/step - loss: 0.0017 - val_loss: 0.0018 - learning_rate: 2.5000e-04

Epoch 25/25
235/235 - 3s - 14ms/step - loss: 0.0017 - val_loss: 0.0018 - learning_rate: 1.2500e-04

17/20 0s 7ms/step

2025-12-07 09:08:54.514077: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_65', 8 bytes spill stores, 8 bytes spill loads

20/20 3s 63ms/step



118/118 1s 5ms/step
20/20 0s 15ms/step
Latent features shapes: (60000, 64) (10000, 64)
Model: "7seg_mlp"

Layer (type)	Output Shape	Param #
input_layer_4 (InputLayer)	(None , 64)	0
dense_8 (Dense)	(None , 128)	8,320
dropout (Dropout)	(None , 128)	0
dense_9 (Dense)	(None , 64)	8,256
dense_10 (Dense)	(None , 7)	455

Total params: 17,031 (66.53 KB)

Trainable params: 17,031 (66.53 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/30

2025-12-07 09:08:57.921304: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:08:57.921359: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:08:57.921403: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:08:57.921416: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:08:57.921426: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:08:58.680162: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_712', 124 bytes spill stores, 124 bytes spill loads

2025-12-07 09:08:59.304624: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_153', 104 bytes spill stores, 104 bytes spill loads

2025-12-07 09:09:00.623786: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_747', 1044 bytes spill stores, 1044 bytes spill loads

2025-12-07 09:09:00.891498: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_714', 64 bytes spill stores, 64 bytes spill loads

2025-12-07 09:09:00.918816: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function

'gemm_fusion_dot_747', 64 bytes spill stores, 64 bytes spill loads

2025-12-07 09:09:01.206031: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_714', 1044 bytes spill stores, 1044 bytes spill loads

2025-12-07 09:09:03.534506: I

external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:03.534571: I

external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:03.534599: I

external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:03.534608: I

external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:03.534616: I

external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:04.182246: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_498', 4 bytes spill stores, 4 bytes spill loads

2025-12-07 09:09:04.217072: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_153', 4 bytes spill stores, 4 bytes spill loads

2025-12-07 09:09:04.389973: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_712', 16 bytes spill stores, 16 bytes spill loads

2025-12-07 09:09:04.837665: I

external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_153', 360 bytes spill stores, 360 bytes spill loads

2025-12-07 09:09:05.501219: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_498', 16 bytes spill stores, 16 bytes spill loads

2025-12-07 09:09:05.600431: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_712', 580 bytes spill stores, 580 bytes spill loads

2025-12-07 09:09:07.315834: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:07.942157: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_59', 116 bytes spill stores, 116 bytes spill loads

235/235 - 12s - 52ms/step - accuracy: 0.2749 - loss: 0.2103 - val_accuracy:
0.3810 - val_loss: 0.0768 - learning_rate: 0.0010

Epoch 2/30

235/235 - 1s - 3ms/step - accuracy: 0.3742 - loss: 0.0762 - val_accuracy: 0.3746
- val_loss: 0.0492 - learning_rate: 0.0010

Epoch 3/30

235/235 - 1s - 4ms/step - accuracy: 0.3728 - loss: 0.0587 - val_accuracy: 0.3740
- val_loss: 0.0434 - learning_rate: 5.0000e-04

2025-12-07 09:09:10.826282: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:10.826358: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:11.618354: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_16', 116 bytes spill stores, 116 bytes spill loads

2025-12-07 09:09:11.691010: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function 'gemm_fusion_dot_9',
104 bytes spill stores, 104 bytes spill loads

1/20 28s 2s/step

2025-12-07 09:09:12.346392: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:12.346447: I
external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All
configs were filtered out because none of them sufficiently match the hints.
Maybe the hints set does not contain a good representative set of valid configs?
Working around this by using the full hints set instead.

2025-12-07 09:09:12.956604: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function 'gemm_fusion_dot_9',
4 bytes spill stores, 4 bytes spill loads

2025-12-07 09:09:13.158955: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_16', 12 bytes spill stores, 12 bytes spill loads

2025-12-07 09:09:13.170500: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_16', 8 bytes spill stores, 8 bytes spill loads

2025-12-07 09:09:13.221539: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_16', 116 bytes spill stores, 116 bytes spill loads

2025-12-07 09:09:13.376621: I
external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas
warning : Registers are spilled to local memory in function 'gemm_fusion_dot_9',
360 bytes spill stores, 360 bytes spill loads

20/20 3s 72ms/step

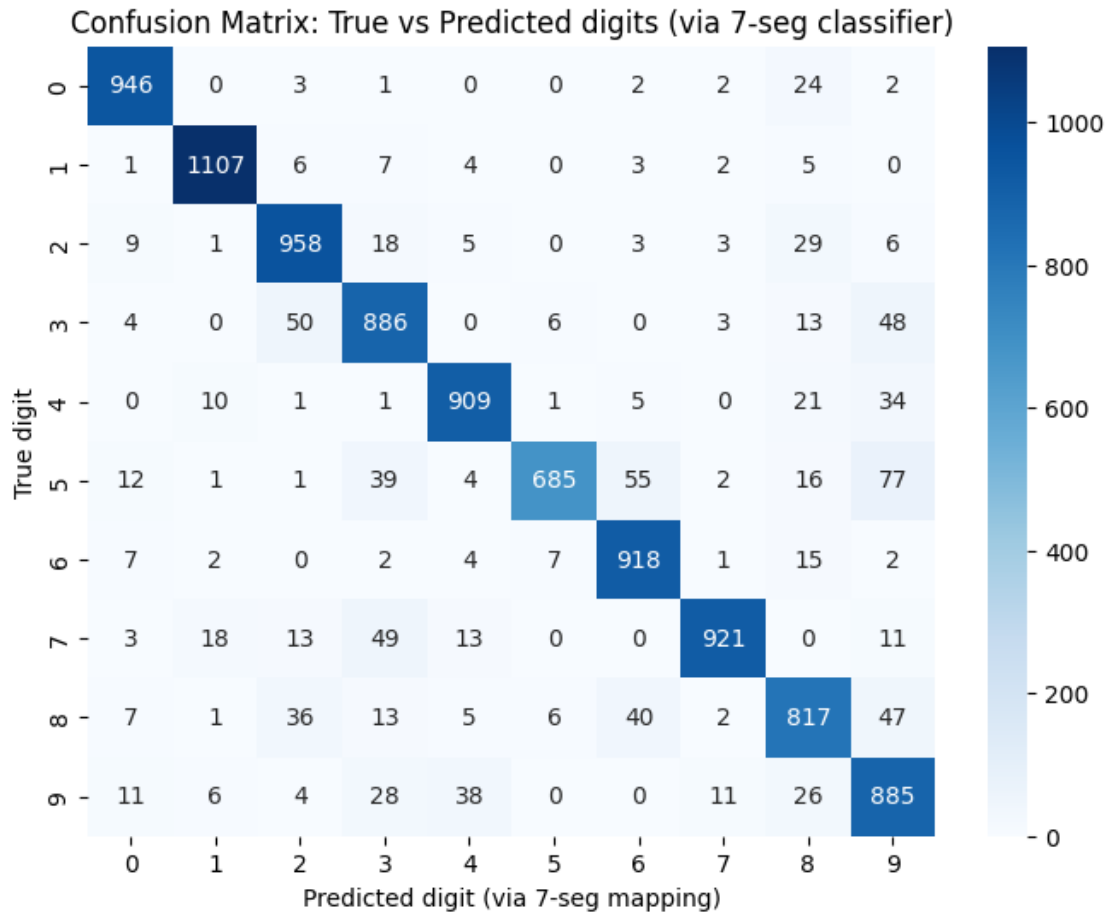
Segment 0 accuracy: 98.44%

Segment 1 accuracy: 97.57%

Segment 2 accuracy: 98.57%

Segment 3 accuracy: 97.33%

Segment 4 accuracy: 96.22%
 Segment 5 accuracy: 96.66%
 Segment 6 accuracy: 97.68%
 Average segment accuracy: 97.50%
 Digit accuracy (mapped from 7-seg predictions): 90.32%



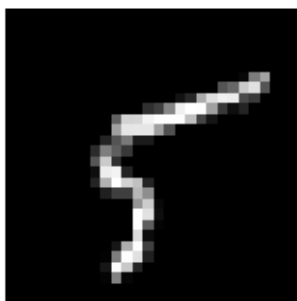
Total mismatches: 968 / 10000
 True: 4 Predicted: 6 Pred seg: [1 0 1 1 1 1 1] Prob seg: [0.786 0.346 0.927
 0.738 0.74 0.994 0.972]

True 4 Pred 6



True: 5 Predicted: 4 Pred seg: [1 1 1 0 0 1 1] Prob seg: [0.662 0.638 0.997
0.315 0.017 0.791 0.65]

True 5 Pred 4



True: 3 Predicted: 2 Pred seg: [1 1 0 1 1 0 1] Prob seg: [1. 0.993 0.358 1.
0.69 0.017 0.999]

True 3 Pred 2



True: 9 Predicted: 3 Pred seg: [1 1 1 1 0 0 0] Prob seg: [0.995 1. 0.998

0.728 0.091 0.184 0.498]

True 9 Pred 3



True: 2 Predicted: 8 Pred seg: [1 1 1 1 1 1 1] Prob seg: [0.999 0.989 0.677
0.994 0.982 0.854 0.978]

True 2 Pred 8



True: 4 Predicted: 8 Pred seg: [1 1 1 1 1 1 1] Prob seg: [0.531 0.945 0.994
0.598 0.686 0.996 0.996]

True 4 Pred 8



True: 7 Predicted: 3 Pred seg: [1 1 1 0 0 0 1] Prob seg: [0.524 1. 1.
0.015 0.001 0.386 0.795]

True 7 Pred 3



True: 0 Predicted: 2 Pred seg: [1 1 1 1 1 0 1] Prob seg: [1. 0.982 0.857
0.999 0.657 0.189 0.891]

True 0 Pred 2

